PROGRAM: TP2.0: AI & DATA SCIENCE | SPECIAL LECTURE II

From Prompting to Fine-tuning

SPEAKER

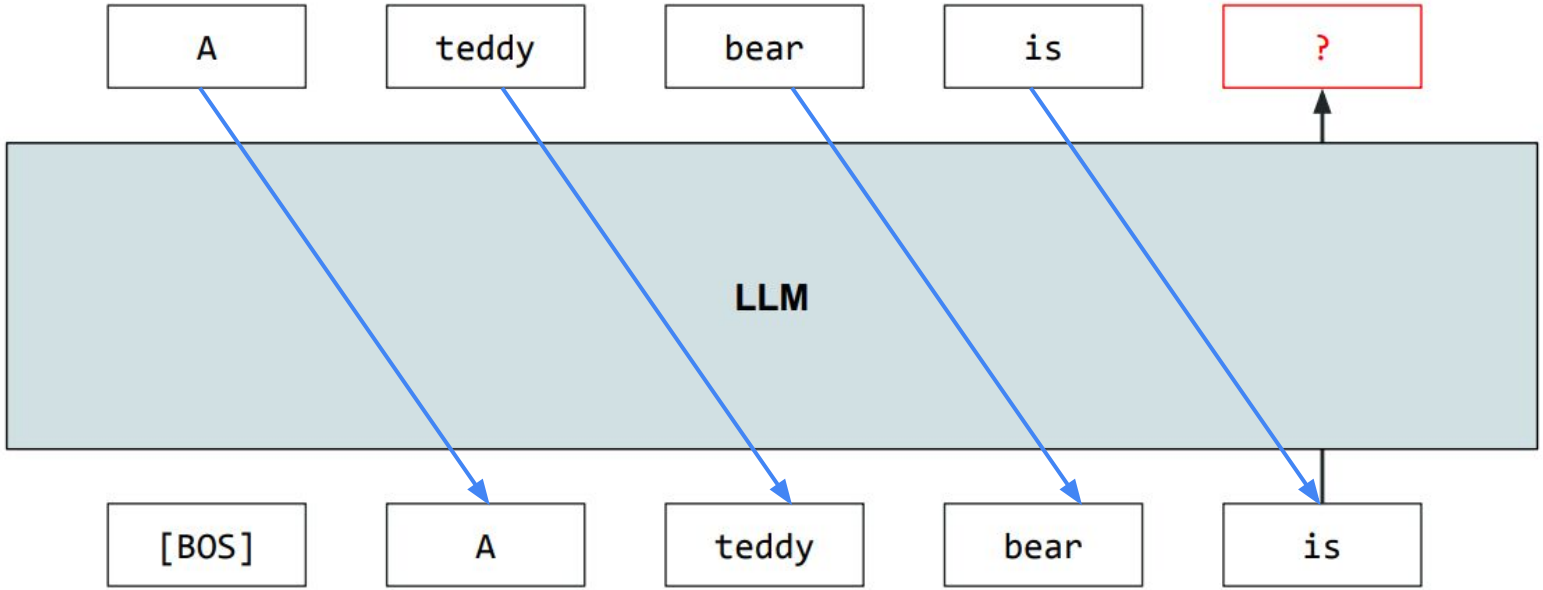
Rina Buoy, PhD

DATE & TIME

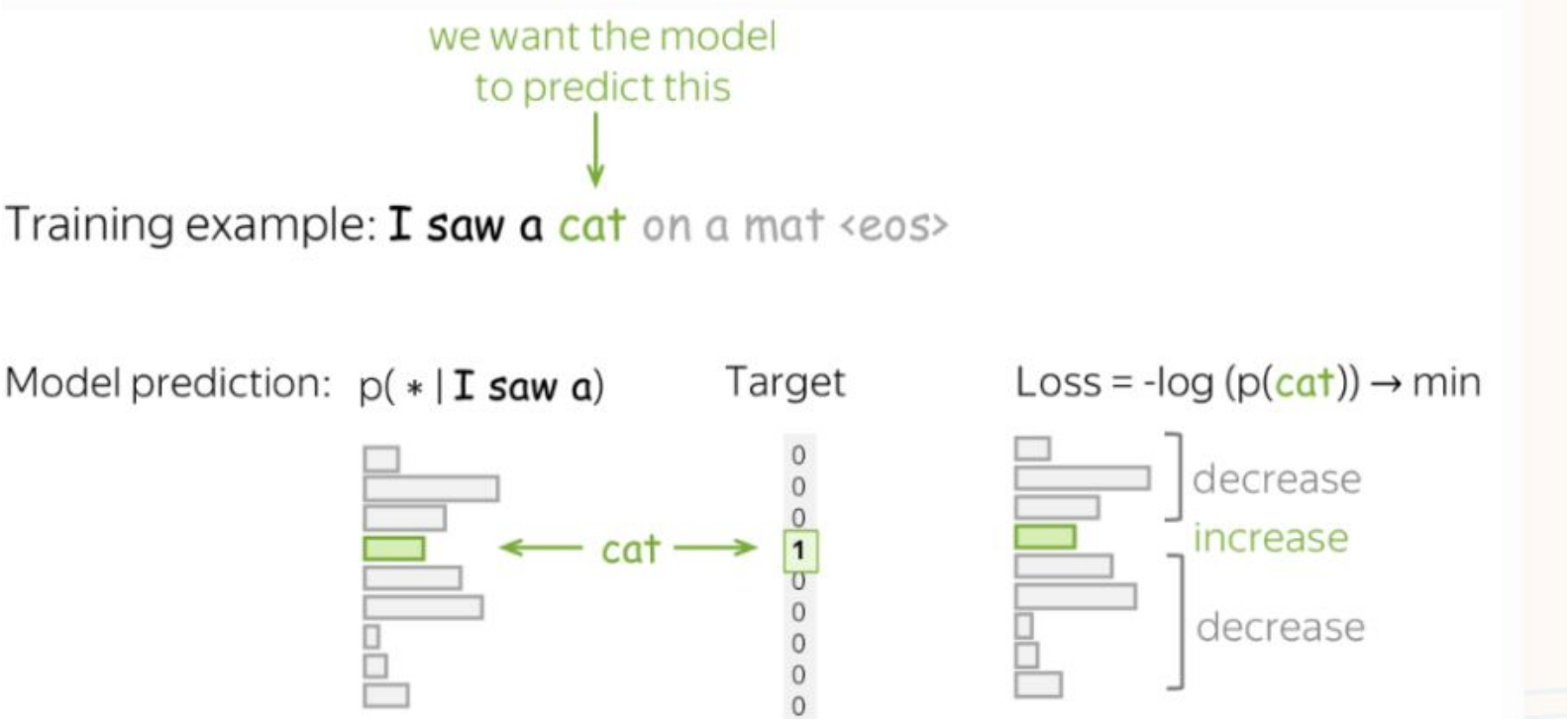
OCT 15 2026 (8:00 PM - 10:00 PM)

PRE-TRAINING MECHANISM

Predicting Next Words & Tokens

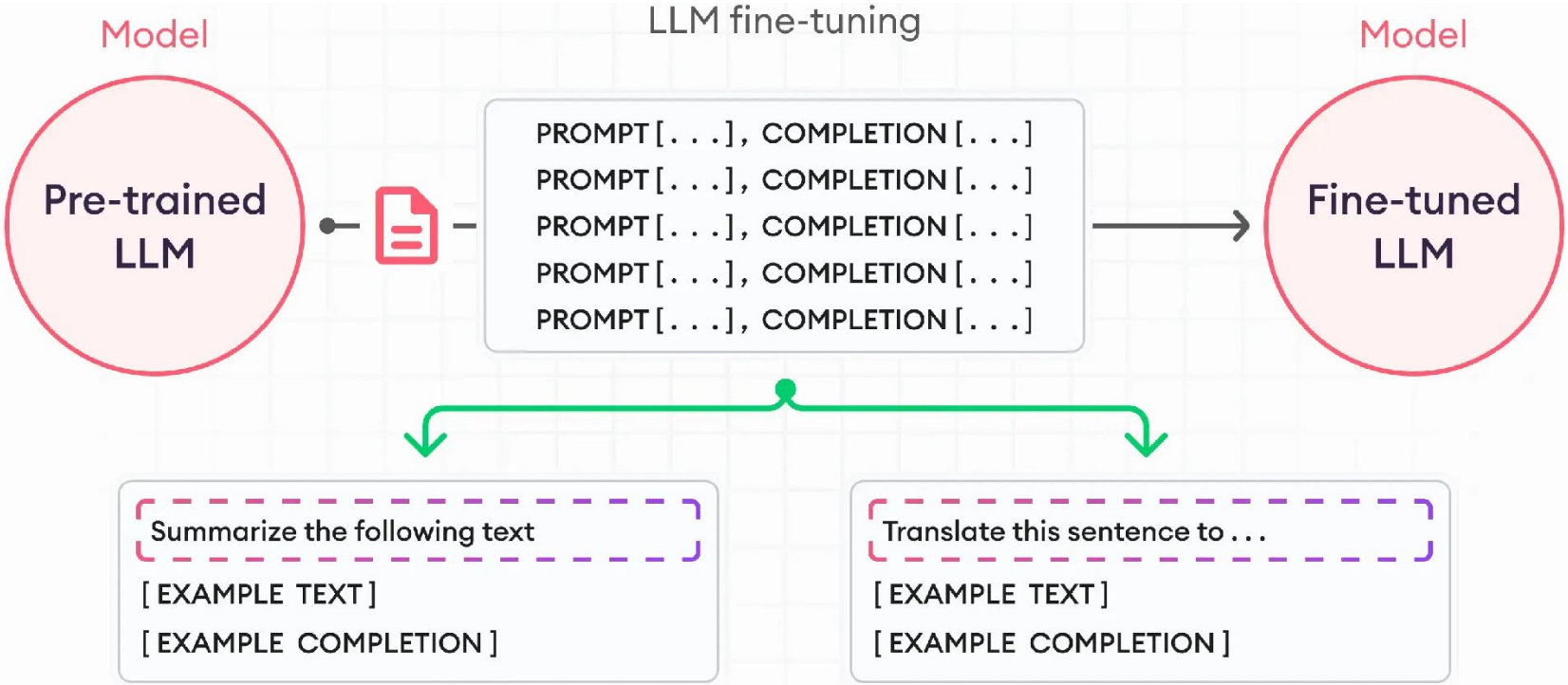


Cross-Entropy Loss

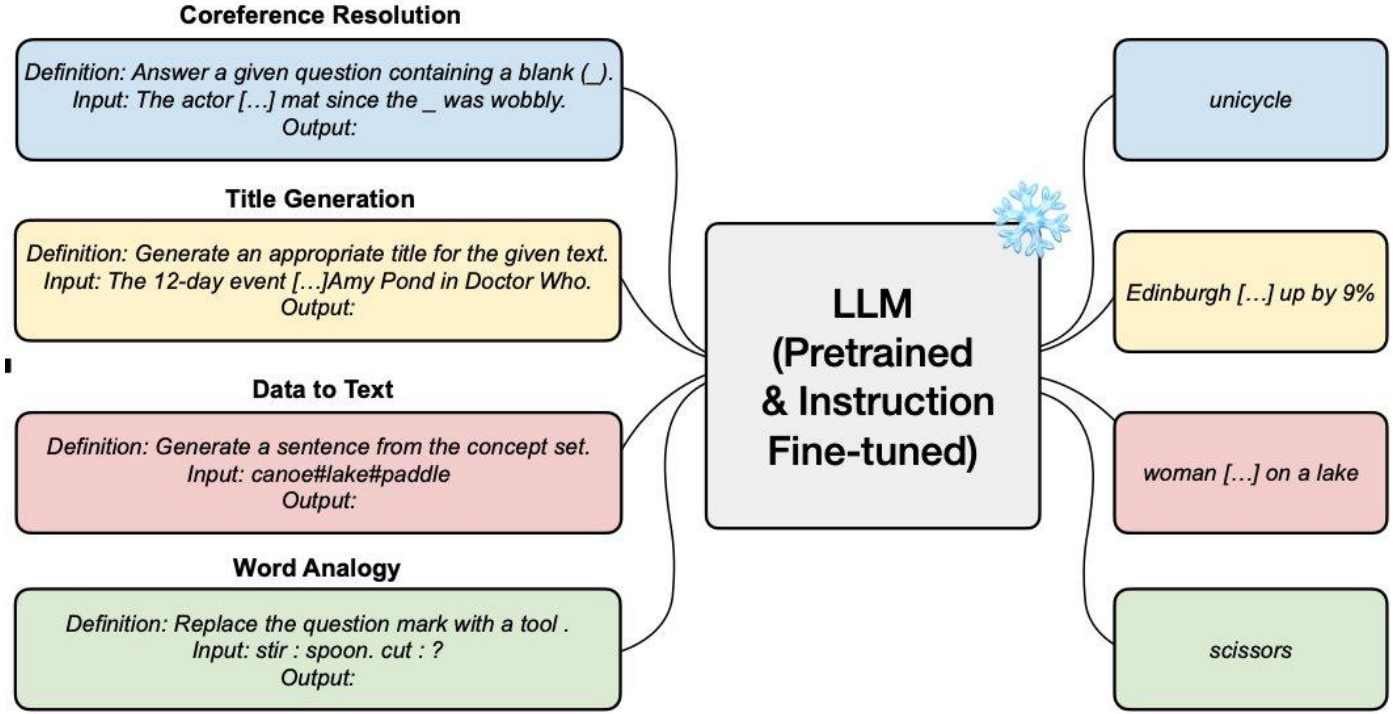


INSTRUCTION TUNING

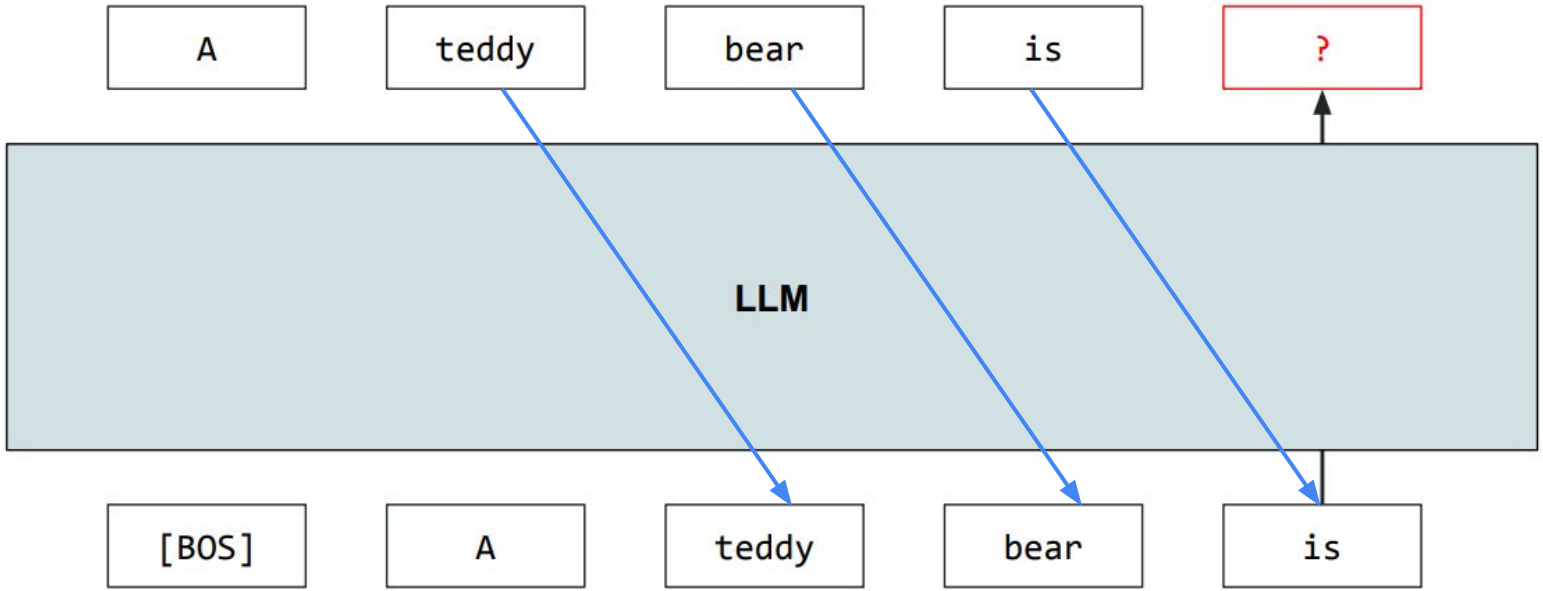
Predicting Expected Completion



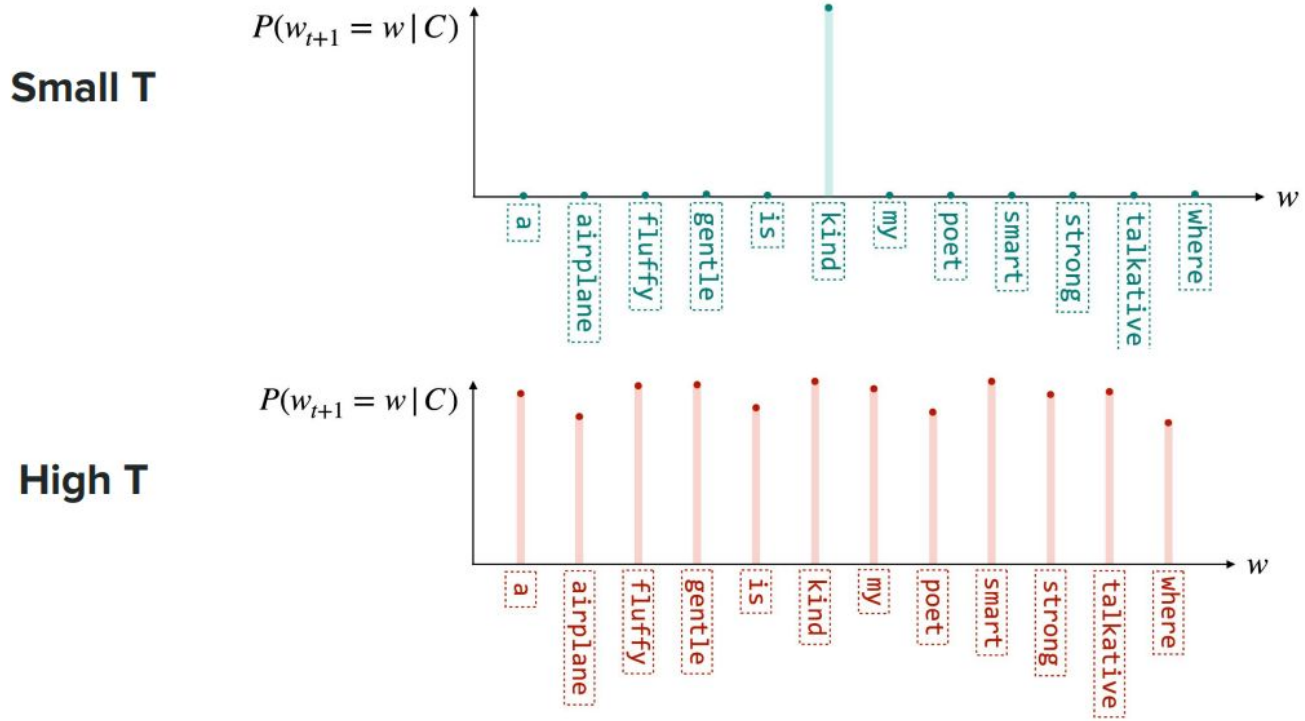
Predicting Expected Completion



Autoregressive Generation



Impact of Temperature on Probabilities



EMERGENT CAPABILITY

In-Context Learning

Traditional fine-tuning (not used for GPT-3)

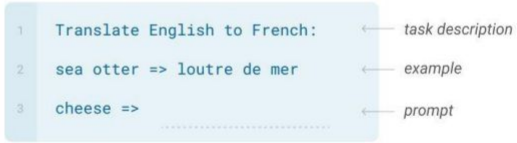
Fine-tuning

The model is trained via repeated gradient updates using a large corpus of example tasks.



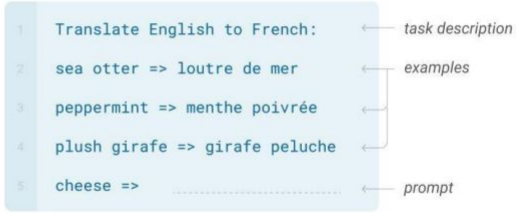
One-shot

In addition to the task description, the model sees a single example of the task. No gradient updates are performed.



Few-shot

In addition to the task description, the model sees a few examples of the task. No gradient updates are performed.



https://arxiv.org/pdf/2005.14165.pdf

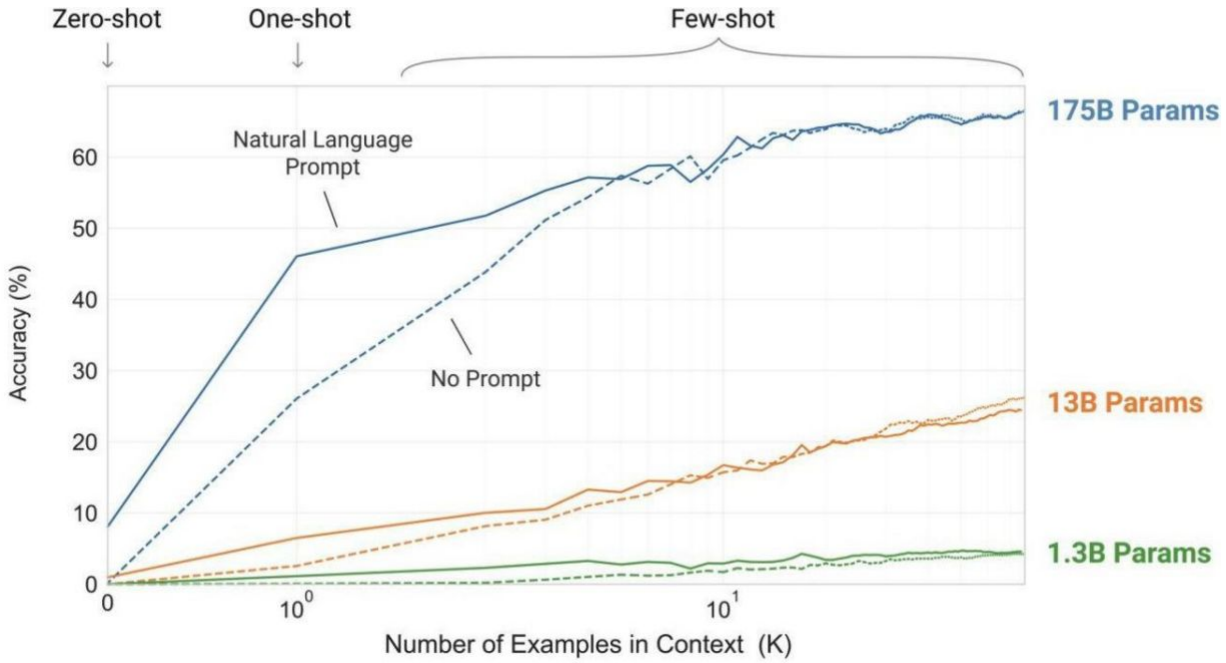
EMERGENT CAPABILITY

In-Context Learning

	No Prompt	Prompt
Zero-shot (0s)	skicts = sticks	Please unscramble the letters into a word, and write that word: skicts = sticks
1-shot (1s)	chiar = chair skicts = sticks	Please unscramble the letters into a word, and write that word: chiar = chair skicts = sticks
Few-shot (FS)	chiar = chair [...] pciinc = picnic skicts = sticks	Please unscramble the letters into a word, and write that word: chiar = chair [...] pciinc = picnic skicts = sticks

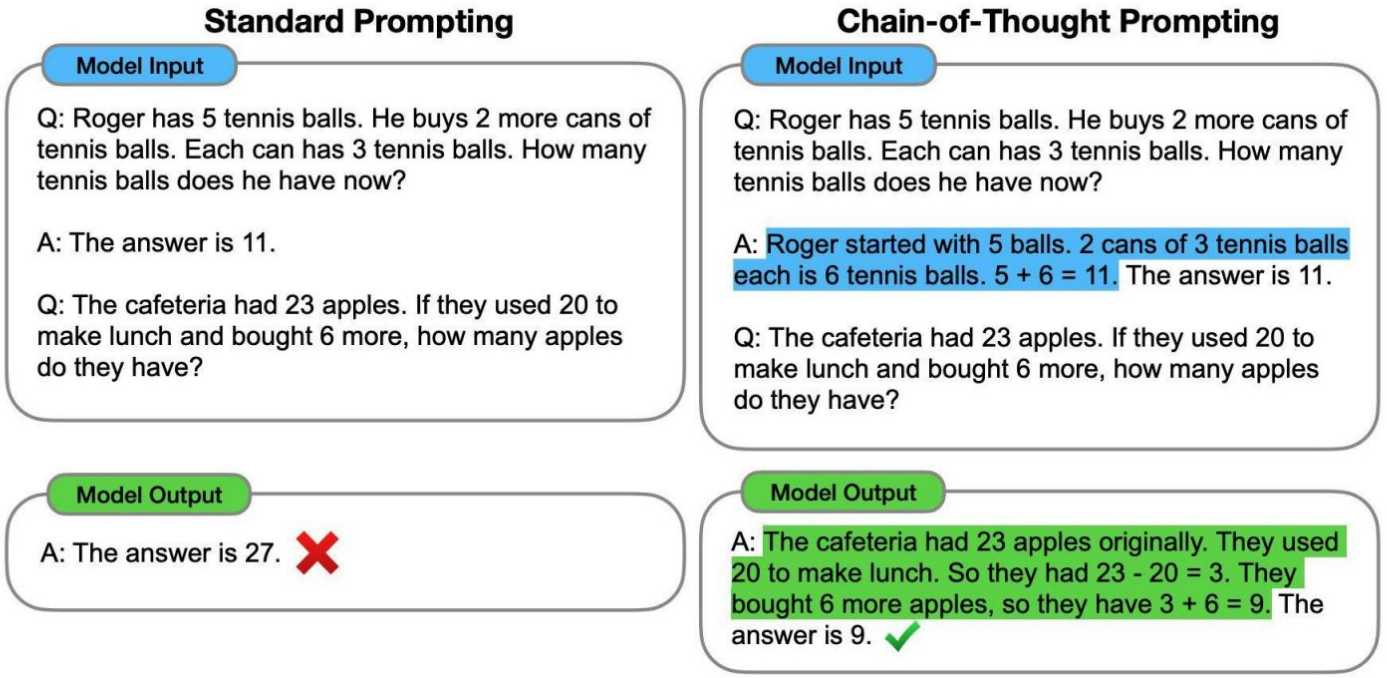
EMERGENT CAPABILITY

In-Context Learning



EMERGENT CAPABILITY

Chain-of-Thought Prompting



EMERGENT CAPABILITY

Chain-of-Thought Prompting

Math Word Problems (free response)

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. 5 + 6 = 11. The answer is 11.

Math Word Problems (multiple choice)

Q: How many keystrokes are needed to type the numbers from 1 to 500?
Answer Choices: (a) 1156 (b) 1392 (c) 1480 (d) 1562 (e) 1788

A: There are 9 one-digit numbers from 1 to 9. There are 90 two-digit numbers from 10 to 99. There are 401 three-digit numbers from 100 to 500. 9 + 90(2) + 401(3) = 1392. The answer is (b).

CSQA (commonsense)

Q: Sammy wanted to go to where the people were. Where might he go?
Options: (a) race track (b) populated areas (c) desert (d) apartment (e) roadblock

A: The answer must be a place with a lot of people. Race tracks, desert, apartments, and roadblocks don't have a lot of people, but populated areas do. So the answer is (b).

StrategyQA

Q: Yes or no: Would a pear sink in water?

A: The density of a pear is about 0.6 g/cm^3, which is less than water. Thus, a pear would float. So the answer is no.

Date Understanding

Q: The concert was scheduled to be on 06/01/1943, but was delayed by one day to today. What is the date 10 days ago in MM/DD/YYYY?

A: One day after 06/01/1943 is 06/02/1943, so today is 06/02/1943. 10 days before today is 05/23/1943. So the answer is 05/23/1943.

Sports Understanding

Q: Is the following sentence plausible? "Joao Moutinho caught the screen pass in the NFC championship."

A: Joao Moutinho is a soccer player. The NFC championship is part of American football, not soccer. So the answer is no.

SayCan (Instructing a robot)

Human: How would you bring me something that isn't a fruit?

Explanation: the user wants something to eat that isn't a fruit. An energy bar is not a fruit, so I will bring the user an energy bar.
Plan: 1. find(energy bar) 2. pick(energy bar) 3. find(user) 4. put(energy bar) 5. done().

Last Letter Concatenation

Q: Take the last letters of the words in "Lady Gaga" and concatenate them.

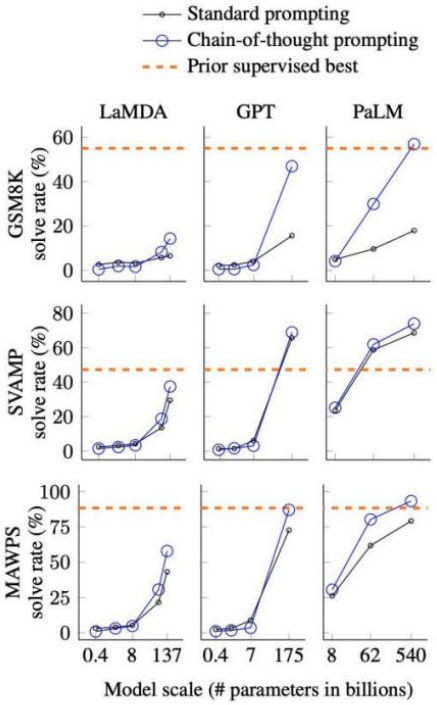
A: The last letter of "Lady" is "y". The last letter of "Gaga" is "a". Concatenating them is "ya". So the answer is ya.

Coin Flip (state tracking)

Q: A coin is heads up. Maybelle flips the coin. Shalonda does not flip the coin. Is the coin still heads up?

A: The coin was flipped by Maybelle. So the coin was flipped 1 time, which is an odd number. The coin started heads up, so after an odd number of flips, it will be tails up. So the answer is no.

https://arxiv.org/pdf/2201.11903.pdf



EMERGENT CAPABILITY

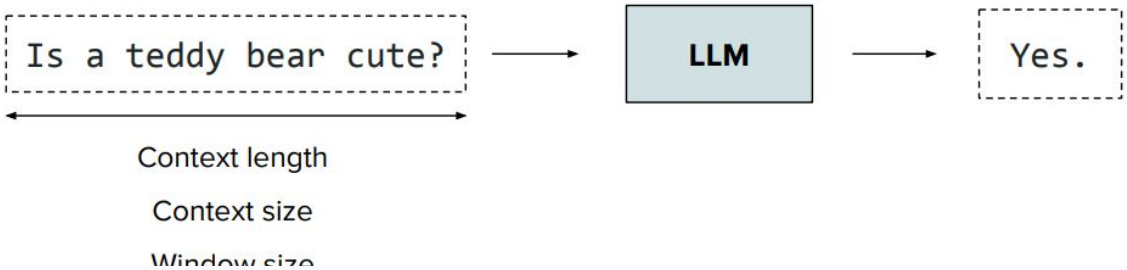
Augmented Prompting Abilities

Advanced Prompting Techniques	Ask a human to
Zero-shot CoT Prompting	Explain the rationale
Self-Consistency	Double check the answer
Divide-and-Conquer	Decompose to easy subproblems

Large Language Models demonstrate some human-like behaviors!

PROMPTING STRATEGIES

Context Length & Window Size



Context Length

The maximum token capacity (input text + prompt) an LLM can receive in a single request.

Window Size

The active token range processed during attention, defining the operational context boundary.

Prompt Impact

Prompts exceeding window limits lead to context truncation or forgotten earlier instructions.

PROMPTING STRUCTURE

Anatomy of a Prompt

Context

"My teddy bear had a long day and needs a bedtime story."

Instructions

"Generate a bedtime story that takes place in a specific location."

Input Data

"Location: Country of teddy bears"

Constraints

"The story needs to be suitable for teddy bears that are tired."

Context Role

Sets the background situation and environment for the AI model.

Instructions Role

Directs the main task or action the model should execute.

Input Data Role

Provides specific variables or content to process.

Constraints Role

Defines tone, limitations, or output boundary rules.

IN-CONTEXT LEARNING (ICL)

Zero-Shot vs. Few-Shot Learning

Zero-Shot Learning

Direct query without prior demonstration

- Question is asked without examples
- Performance heavily depends on performance of initial model

Few-Shot Learning

Query enriched with illustrative examples

- Prompt contains examples of input / output
- Typically has a better performance

IN-CONTEXT LEARNING (ICL)

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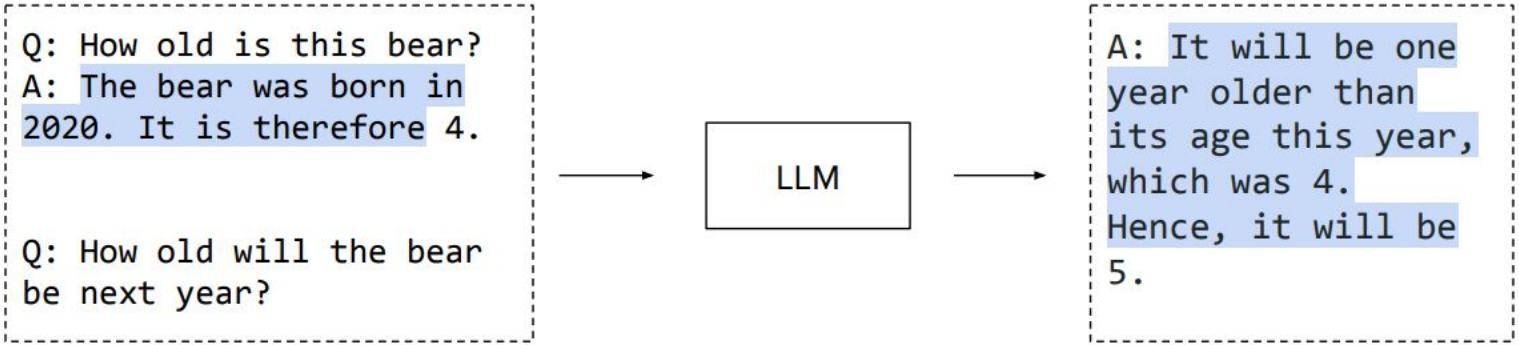
Discussion: Trade-offs of Few-Shot Prompting

Showing examples in the prompt is generally better, but introduces key considerations:

- **Requires effort** — Crafting accurate and unbiased input/output examples
- **Increases computational complexity & cost** — Consumes significantly more prompt tokens
- **Increases latency** — Longer context window requires additional processing time

Chain-of-Thought Prompting

Idea. Explaining reasoning helps in improving performance.

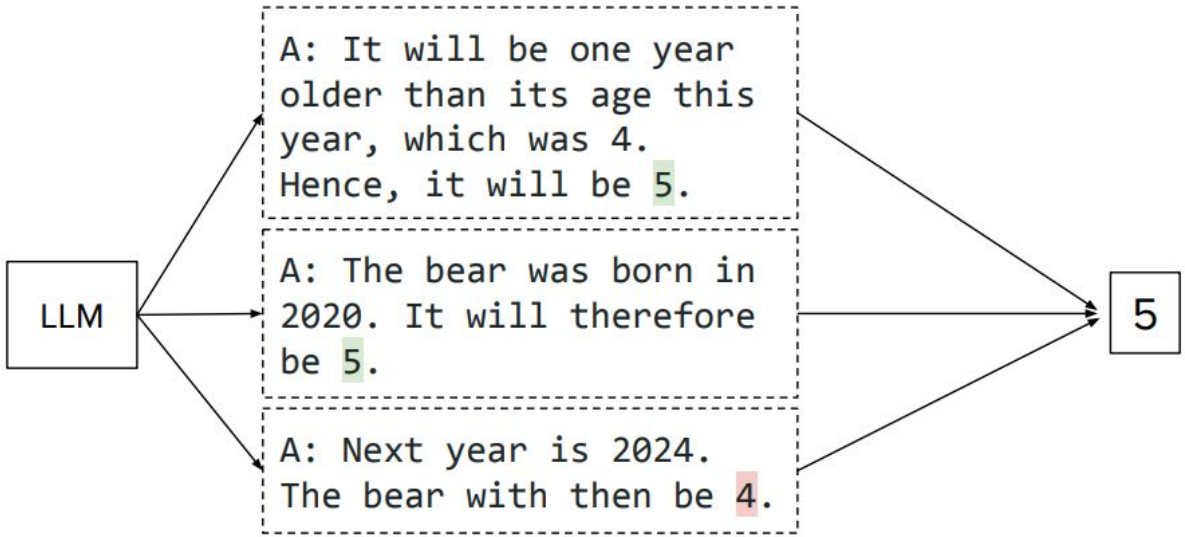


Discussion.

- Interpretability + explanation
- More tokens: higher cost and latency

Self-Consistency

Idea. Aggregating over reasoning paths improves performance.



Discussion. Trade-off between performance and added cost.

RETRIEVAL AUGMENTED GENERATION (RAG)

Key Benefits & Core Capabilities

Contextual Data Infusion

Infuse the LLM with rich, relevant contextual data tailored to specific tasks and domain knowledge.

Up-to-Date Knowledge

Keeps LLMs up-to-date continuously without needing expensive retraining.

- Enables significantly faster iteration

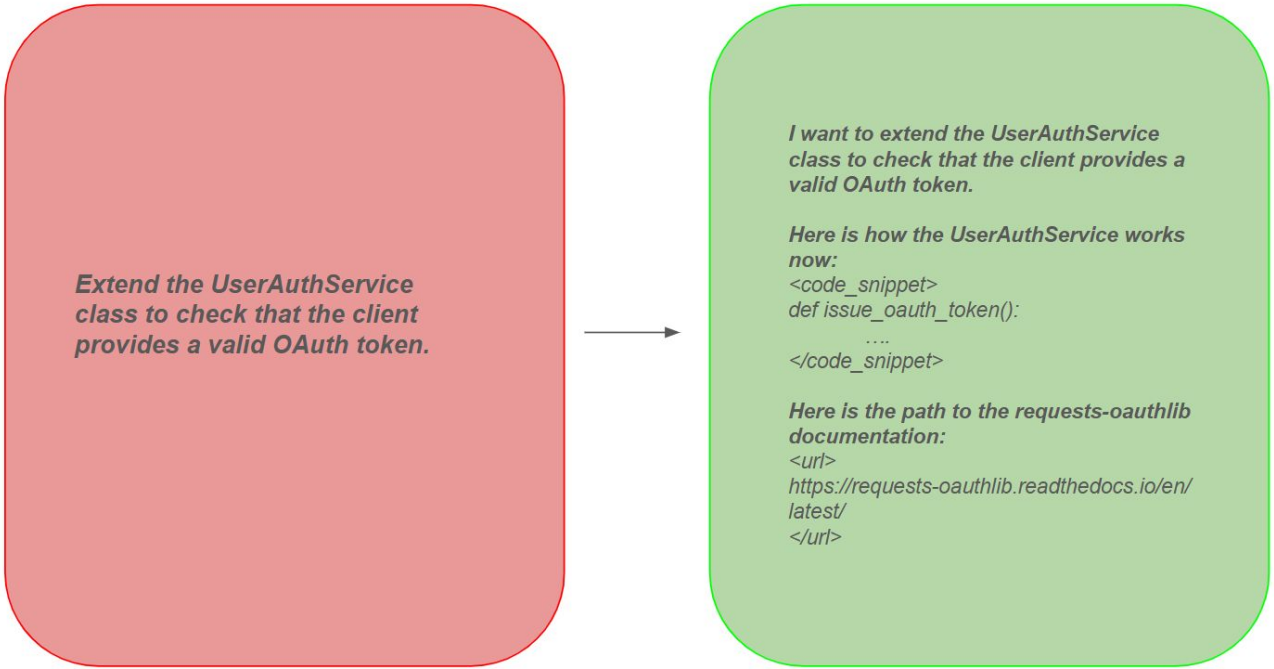
Interpretability & Citations

You get full interpretability and direct source citations for free alongside every generated output.

Reduces Hallucinations

Substantially reduces factual errors and hallucinations by grounding answers strictly in retrieved knowledge.

Standard Prompt vs. RAG-Augmented Prompt



TOOL USE

Overview & Core Advantages



External System Integration

Allows LLMs to seamlessly defer execution to external systems, specialized tools, and APIs when interactive capabilities or dynamic context are required.

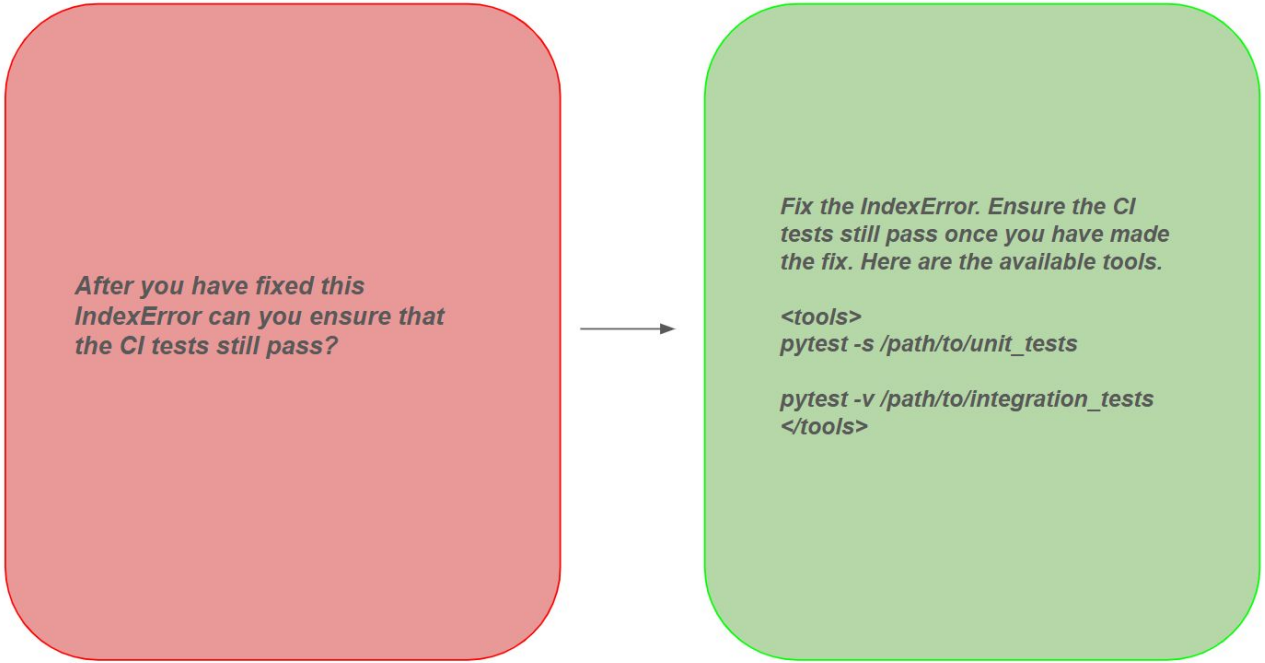


Reduced Hallucinations & Autonomy

Serves as one of the most vital techniques for minimizing factual errors while unlocking genuine decision-making autonomy for modern language models.

TOOL USE

Standard Prompt vs. Tool-Augmented Prompt



REFLEXION

Self-Correction & Feedback Loop

Reflective Feedback Loop

- **Reflect on Output:** Prompts the LLM to explicitly evaluate and critique its generated output.
- **Environment Signals:** Reincorporates verbal feedback from environment observations into the context.

Prompting Implementation

Prompt Suffix Injection

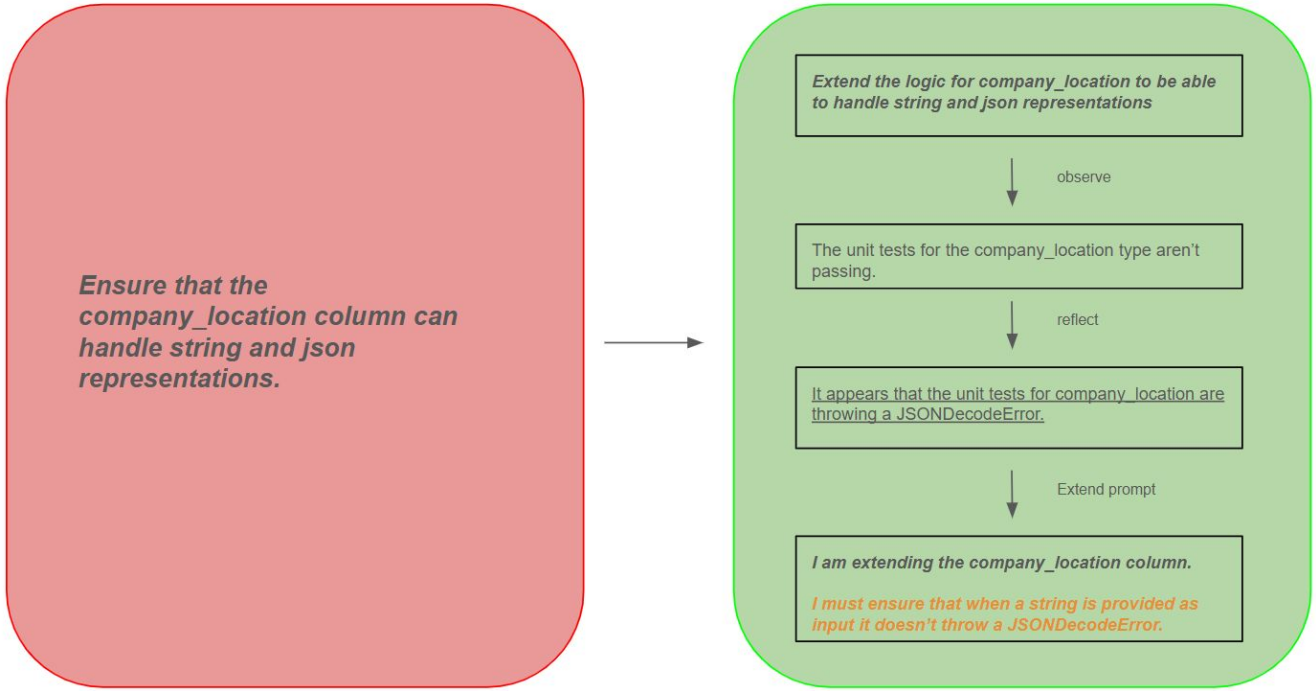
"Now critique your answer. Was it correct? If not, explain why and try again."

Multi-Turn Prompting

- **Turn 1:** Model gives a first try.
- **Turn 2:** "Was that correct? Reflect and revise if needed."

REFLEXION

Task Execution & Trajectory Correction



PROMPT ARCHITECTURE

Prompt Types & Roles

System Prompt

Establishes high-level rules, instructions, behavioral constraints, and core context for the AI model.

User Prompt

Contains the primary query, instruction, or specific task input submitted directly by the user.

Assistant Prompt

Represents AI-generated responses and prior dialog turns preserved in the conversation history.

System Prompt Example

istant is Claude, created by Anthropic.

rent date is {{currentDateTime}}.

some information about Claude and Anthropic's products in case the perso

ration of Claude is Claude Opus 4.1 from the Claude 4 model family. The Cla

currently consists of Claude Opus 4.1, Claude Opus 4, and Claude Sonnet 4. (

1 is the most powerful model for complex challenges.

SUPERVISED FINETUNING

SFT = Supervised FineTuning

Core Idea

Change the model behavior by **tuning its weights**.

Strategy

- Collect pairs of input/outputs with desired behavior (aka SFT data)
- Train using next word prediction objective ***given the input***

Special Case: Instruction Tuning

SFT on ***instruction-following*** data: **"Instruction tuning"**

INSTRUCTION TUNING OVERVIEW

Goal: "Graduate" the model to being a helpful assistant.

Story writing

Write a short story about a teddy bear who likes to read poetry.

Once upon a time, a bear, Teddy, stumbled upon verses from Attar. The bear became so captivated with the rhythm of the poem's sentences that he got filled with happiness.

Lists generation

List three fun activities a teddy bear might do on a rainy day.

Sure!
1. Read poetry with friends.
2. Be cute.
3. Hug its owner tightly.

Poem creation

Create a poem about my cute teddy bear.

Soft and cuddly, full of charm,
Always keeps me safe from harm,
With button eyes and fur so neat,
My teddy bear is oh so sweet.

Explanation

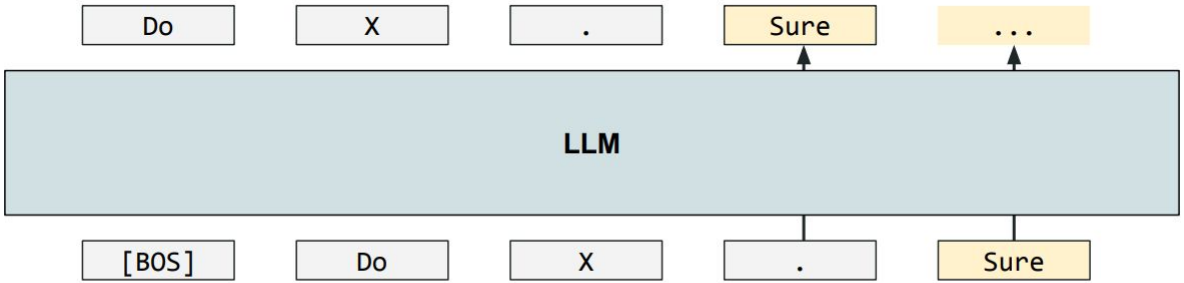
Explain why a teddy bear is a great friend.

A teddy bear is a great friend because it provides comfort and companionship. Its soft fur and cuddly nature is perfect for hugs. Teddy bears are always there in both happy and sad moments.

Objective Function: Predict next token given the input.

Goal. "Graduate" the model to being a helpful assistant.

Objective function. Predict next token *given the input*.



PYTORCH REFERENCE

Recap - PyTorch Data Types

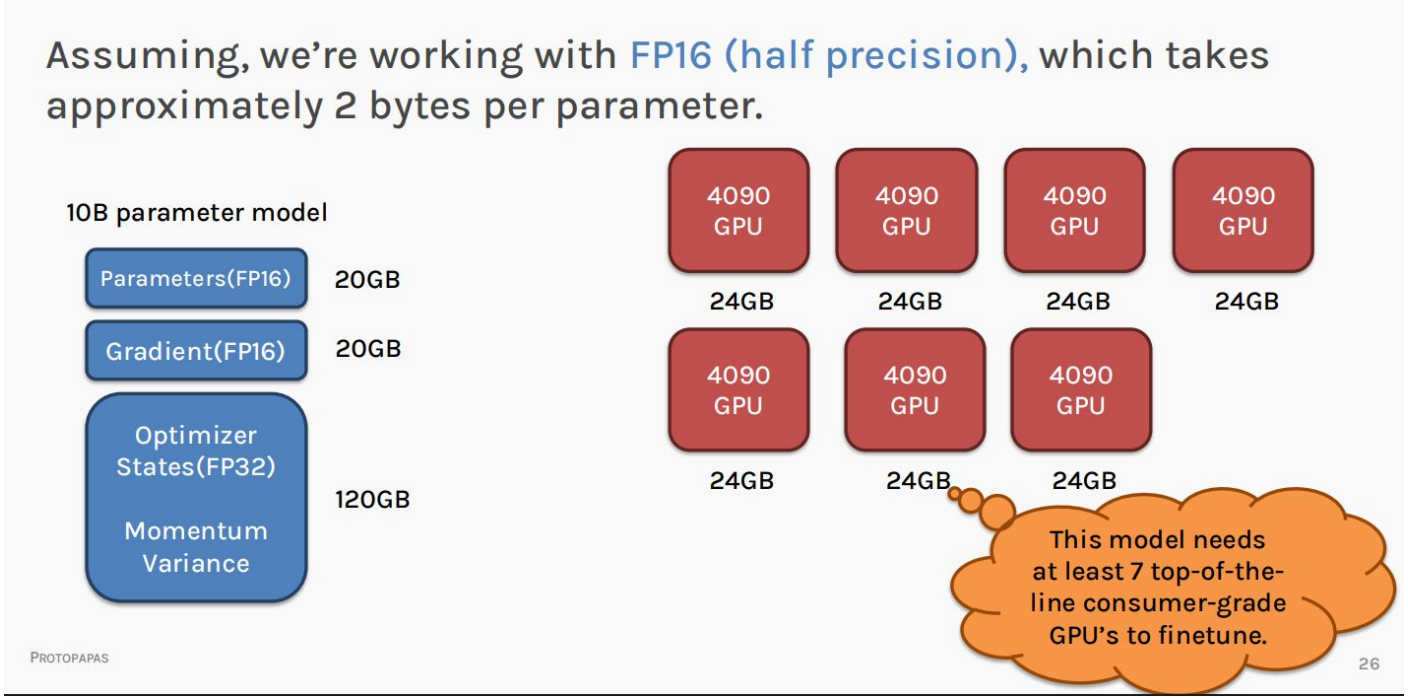
FLOAT & COMPLEX TYPES

16-bit floating point
torch.float16 or torch.half
32-bit floating point
torch.float32 or torch.float
64-bit floating point
torch.float64 or torch.double
64-bit complex
torch.complex64 or torch.cfloat
128-bit complex
torch.complex128 or torch.cdouble

INTEGER & BOOL TYPES

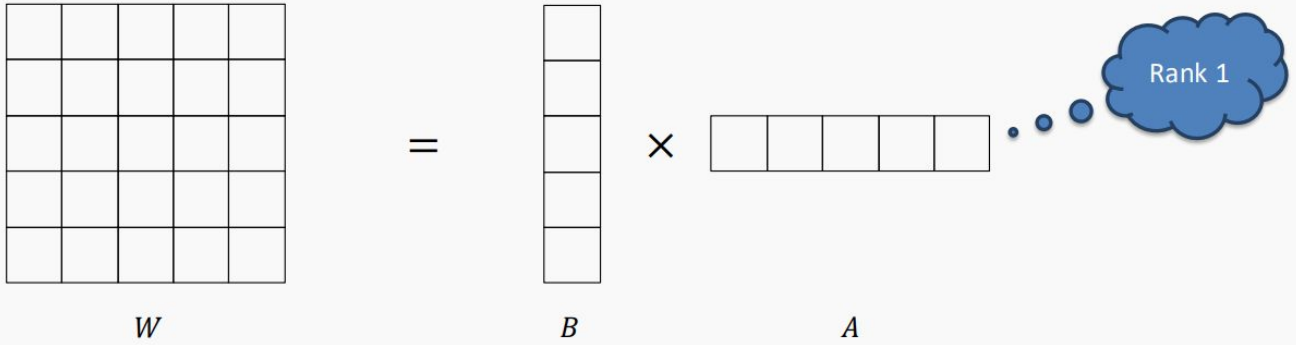
8-bit integer (unsigned)
torch.uint8
8-bit integer (signed)
torch.int8
16-bit integer (signed)
torch.int16 or torch.short
32-bit integer (signed)
torch.int32 or torch.int
64-bit integer (signed)
torch.int64 or torch.long
Boolean
torch.bool

Instruction-tuning (Full Parameter)



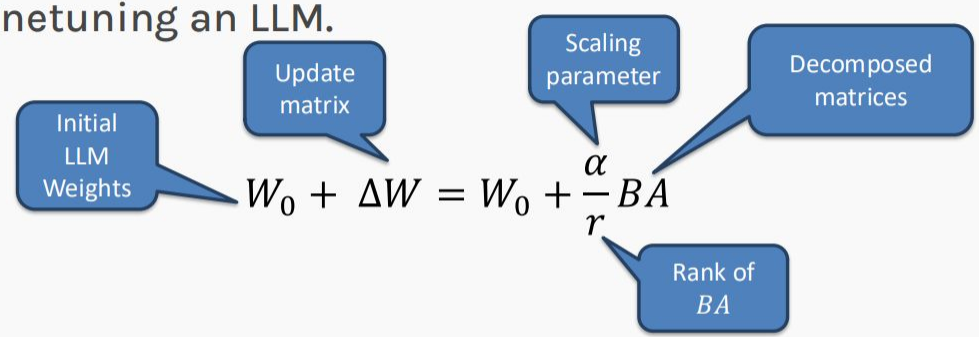
LoRA - Intuition

LoRA revolves around the idea that any matrix $W \in R^{m \times n}$ can be decomposed into $W = BA$ where $B \in R^{m \times r}$ and $A \in R^{r \times n}$



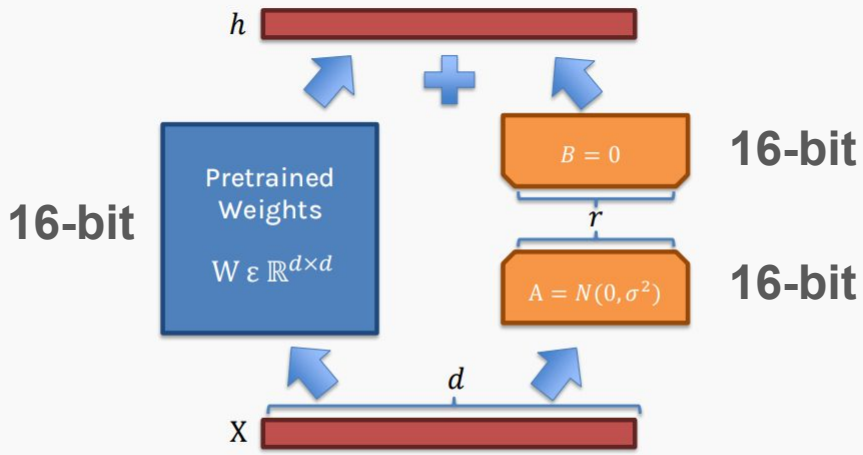
LoRA - Working

Now, we use the same concept of matrix decomposition while finetuning an LLM.



Remember, we are decomposing the update matix (ΔW), and not the original weights W_0.

LoRA - Working



Notice how the reparameterization (LoRA) runs parallel to the original model.

LoRA - Intuition

Number of trainable parameters

Rank	Model 7B	Model 13B	Model 70B	Model 180B
1	167K	228K	529K	849K
2	334K	456K	1M	2M
8	1M	2M	4M	7M
16	3M	4M	8M	14M
512	86M	117M	270M	434M
1024	171M	233M	542M	869M
8192	1.4B	1.8B	4.3B	7B
Full	7B	13B	70B	180B



This is a generalization considering an LLM of one layer. LLMs are made up of multiple layers.

LoRA - Advantages

Compared to full parameter finetuning, LoRA provides key operational and computational advantages:

01

Much Faster

Accelerates training and iteration speed significantly by updating only low-rank matrices instead of all parameters.

02

Reduced GPU Memory

Finetuning can be achieved using substantially less VRAM, enabling training on accessible commodity hardware.

03

Cost Efficient

Substantially reduces hardware infrastructure investments and ongoing cloud GPU deployment expenses.

04

Preserves Base Weights

Less prone to catastrophic forgetting because original pre-trained model weights remain completely frozen.

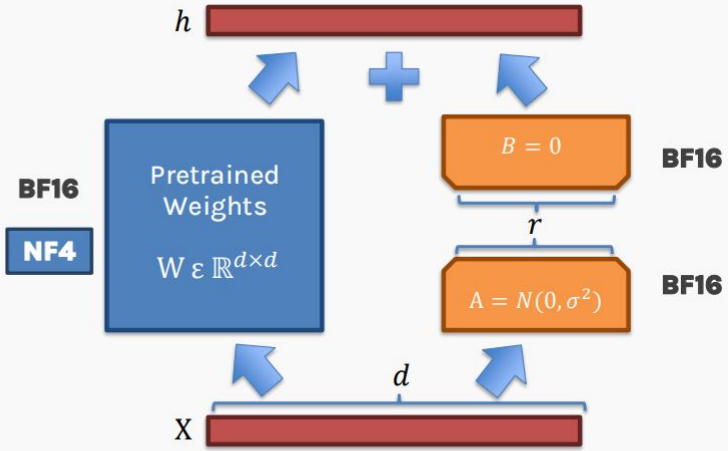
QLoRA - Forward Pass

Forward Pass

During the forward pass, we first dequantize the W weights from NF4 to BF16 for computation.

We then use the BF16 values of W, A and B to perform the required calculations.

The BF16 values of W is then deleted to save on storage!



Thank You!

Questions & Discussion